

Passive RLC Band-Pass Filter Design

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Abstract

This paper will describe the calculations of creating a passive RLC band-pass filter. One of these poles will be at 28 MHz, and the other pole will be at 28.1 MHz, making the passband span 0.1 MHz. This is primarily designed for the CW portion of the 10 meter band, and this would be designed in collaboration with the PATRIK (RPI RCOS) project.

1 Introduction

In order to minimize the amount of spurious emissions produced from the transmitter, we must use a filter to remove the 2nd and 3rd order harmonics. This is not only to clean up the output of the transmitter, but also to comply with FCC guidelines if the PATRIK radio eventually becomes a flushed-out product. As said before, the pass-band will be 0.1 MHz and span from 28 MHz to 28.1 MHz.

2 Filter Calculations

2.1 Initial Frequency Choices

Let : $f_1 = 28 \cdot 10^6$ Hz, and $f_2 = 28.1 \cdot 10^6$ Hz.

Since this filter will be passive, the best we can do (with only one filter) is $+20$ dB/dec and -20 dB/dec on both sides of the passband.

Of course, we also want the passband to be 0 dB/dec in order to lessen the amount of attenuation on the signal we want to be actually transmitted.

2.2 Initial Calculations

In order to get the right values when calculating, we must convert the Hz frequencies into $\frac{rad}{s}$ frequencies. Thus, we must multiply each frequency by 2π .

$$p_1 = (28 \cdot 10^6)(2\pi) = 1.7593 \cdot 10^8 \frac{rad}{s}$$

$$p_2 = (28.1 \cdot 10^6)(2\pi) = 1.7656 \cdot 10^8 \frac{rad}{s}$$

These values will be used to actually calculate the component values.

Next, referring to the bandpass filter equation:

$$H(s) = \frac{2\alpha s}{s^2 + 2\alpha s + \omega_0^2}$$

Given that we already have two poles, the denominator must be $(s + 1.7593 \cdot 10^8)(s + 1.7656 \cdot 10^8)$, so we get:

$$H(s) = \frac{2\alpha s}{(s + 1.7593 \cdot 10^8)(s + 1.7656 \cdot 10^8)}$$

Now, we must expand the denominator in order to find α and ω_0 :

$$H(s) = \frac{2\alpha s}{s^2 + 3.5249 \cdot 10^8 s + 3.106 \cdot 10^{16}}$$

So, we get the values

$$2\alpha = 3.5249 \cdot 10^8 \text{ and } \omega_0^2 = 3.106 \cdot 10^{16}$$

2.3 Component Values

Using these values, we can now calculate the values for the components in the RLC circuit! Using the formulas:

$$2\alpha = \frac{R}{L} \text{ and } \omega_0^2 = \frac{1}{LC}$$

$$3.5249 \cdot 10^8 = \frac{R}{L}$$

$$3.106 \cdot 10^{16} = \frac{1}{LC}$$

Let's choose $L = 1 \mu H$. So, plugging this into the equations, we get:

$$R = 352.5 \Omega$$

$$C = 32.19 pF$$

Therefore, these are the values we will use in the band-pass filter!